

Teacher targets · offline

GT video clip



Action-aligned window



Track4World



Geometry
1024D



2D Motion
128D



3D Motion
256D



Camera
18D



Visibility
4D

Aggregation + per-block L₂ norm



Teacher cache

4D-aware
teacher descriptor



r_t^T

MSE

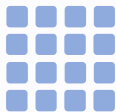
Student branch

30 ×
World-Action MoT

Mid layer 15



16 learnable queries



2-layer Transformer decoder

Self
attn

Q

Cross
attn

FFN



Pool
&
MLP

Student descriptor

r_t^S